

COLLEGE CODERS

React JS Lesson 2 – JSX and Dynamic Data (Premium Notes)

1) What JSX is and why React uses it

- JSX stands for JavaScript XML.
- It allows writing HTML-like syntax inside JavaScript.
- React uses JSX because UI becomes easy to read and structure.

Example:

```
function App(){  
  return <h1>Hello React</h1>  
}
```

Explanation:

- Here h1 looks like HTML but actually JSX.
- React converts JSX into JavaScript using Babel.
- That converted code becomes `React.createElement()`.

2) JavaScript inside JSX using { }

- Curly braces allow JavaScript expressions inside JSX.
- Variables, calculations, and function calls can be written.

Example:

```
const name = "Rohith";  
  
function App(){  
  return <h1>Hello {name}</h1>  
}
```

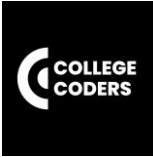
Explanation:

- name variable value appears on screen.
- Only expressions work inside braces.

3) Variables, functions, dynamic values on screen

- Variables store values.
- Functions return values dynamically.

Example:



COLLEGE CODERS

```
const age = 22;
```

```
function getCountry(){  
  return "India";  
}
```

```
function App(){  
  return <h2>{age} - {getCountry()}</h2>  
}
```

Explanation:

- age prints directly.
- Function executes and returns India.

4) Profile Card demo

- Multiple values can be displayed together.

Example:

```
const name = "Rohith";  
const age = 22;  
const country = "India";
```

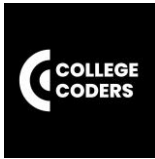
```
function App(){  
  return (  
    <div>  
      <h2>{name}</h2>  
      <p>{age}</p>  
      <p>{country}</p>  
    </div>  
  )  
}
```

Explanation:

- One card displays multiple dynamic values.

5) JSX Rules & Best Practices

- Must return one parent element.
- Use className instead of class.
- Use camelCase attributes.



COLLEGE CODERS

- Close all tags.
- Only expressions allowed.

Example:

```
return (  
  <div className="box">  
      
  </div>  
)
```

Explanation:

- div is parent wrapper.
- className replaces class.
- img tag must close.

6) Dynamic data examples

- Time and math can be shown directly.

Example:

```
const user = "Student";  
  
function App(){  
  return (  
    <div>  
      <h2>{user}</h2>  
      <p>{2+3}</p>  
      <p>{new Date().toLocaleTimeString()}</p>  
    </div>  
  )  
}
```

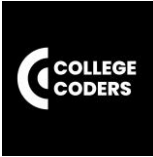
Explanation:

- Math result appears instantly.
- Current time updates on render.

7) Mini Project – Profile Card

- Use variables and style card.

Example:



COLLEGE CODERS

```
const name="Rohith";  
const age=22;  
const country="India";
```

```
function App(){  
  return (  
    <div style={{border:'1px solid black',padding:'20px'}}>  
      <h2>{name}</h2>  
      <p>{age}</p>  
      <p>{country}</p>  
    </div>  
  )  
}
```

Explanation:

- Inline style uses object syntax.

8) Assignment

- Create profile card with 4 details.
- Add inline styling.
- Use one extra dynamic value.